G Model coherence evaluation

G.1 Language model evaluation

Table 8: Model evaluations. For each evaluation, we report the orthogonalized model's performance, followed by the baseline model's performance, followed by the absolute increase or decrease.

Chat model	MMLU	TinyHellaSwag	ARC	WinoGrande	GSM8K	TruthfulQA
Qwen 1.8B	43.0 / 43.1 (-0.1)	48.2 / 49.3 (-1.1)	37.6 / 38.7 (-1.1)	59.6 / 59.0 (+0.6)	29.7 / 30.0 (-0.3)	37.1 / 41.7 (-4.6)
Qwen 7B	54.8 / 56.8 (-2.0)	76.3 / 73.1 (+3.2)	52.0 / 51.7 (+0.3)	72.0 / 72.5 (-0.5)	41.8 / 48.1 (-6.3)	47.9 / 51.6 (-3.7)
Qwen 14B	66.1 / 65.9 (+0.2)	77.3 / 79.5 (-2.2)	60.3 / 61.3 (-1.0)	74.8 / 74.7 (+0.1)	59.3 / 60.3 (-1.0)	50.4 / 52.9 (-2.5)
Qwen 72B	76.5 / 77.2 (-0.7)	86.5 / 85.3 (+1.2)	67.2 / 67.6 (-0.4)	80.7 / 80.8 (-0.1)	76.3 / 75.5 (+0.8)	55.0 / 56.4 (-1.4)
Yi 6B	62.6 / 63.2 (-0.6)	78.1 / 76.8 (+1.3)	56.6 / 57.4 (-0.8)	72.9 / 72.2 (+0.7)	39.0 / 40.6 (-1.6)	44.2 / 50.1 (-5.9)
Yi 34B	73.5 / 74.9 (-1.4)	83.6 / 84.6 (-1.0)	65.6 / 64.9 (+0.7)	78.9 / 80.2 (-1.3)	65.5 / 65.0 (+0.5)	51.9 / 55.4 (-3.5)
Gemma 2B	36.8 / 36.9 (-0.1)	57.1 / 55.2 (+1.9)	43.0 / 43.3 (-0.3)	60.5 / 61.5 (-1.0)	10.8 / 11.1 (-0.3)	40.4 / 45.8 (-5.4)
Gemma 7B	51.8 / 51.7 (+0.1)	46.5 / 44.9 (+1.6)	51.7 / 51.5 (+0.2)	66.6 / 66.5 (+0.1)	31.3 / 32.0 (-0.7)	44.7 / 47.1 (-2.4)
Llama-2 7B	46.8 / 47.5 (-0.7)	76.8 / 77.6 (-0.8)	53.0 / 53.7 (-0.7)	71.7 / 72.6 (-0.9)	22.7 / 23.1 (-0.4)	41.6 / 45.3 (-3.7)
Llama-2 13B	53.6 / 53.6 (+0.0)	82.3 / 83.2 (-0.9)	60.4 / 60.3 (+0.1)	73.4 / 74.3 (-0.9)	35.3 / 35.6 (-0.3)	42.6 / 44.0 (-1.4)
Llama-2 70B	63.1 / 63.0 (+0.1)	84.8 / 84.8 (+0.0)	65.2 / 65.4 (-0.2)	79.7 / 80.2 (-0.5)	54.5 / 53.0 (+1.5)	51.8 / 52.8 (-1.0)
Llama-3 8B	65.0 / 65.8 (-0.8)	79.6 / 82.1 (-2.5)	62.3 / 62.4 (-0.1)	75.9 / 75.5 (+0.4)	74.3 / 75.9 (-1.6)	48.3 / 51.7 (-3.4)
Llama-3 70B	79.8 / 79.9 (-0.1)	85.4 / 86.1 (-0.7)	71.5 / 71.8 (-0.3)	83.4 / 83.6 (-0.2)	90.8 / 91.2 (-0.4)	59.5 / 61.8 (-2.3)

Except on TruthfulQA, orthogonalization has a very small effect on general performance benchmarks. We observe less than 1% performance drop on average, with the difference to the baseline performance being indistinguishable from noise in most cases. The main exceptions are Qwen 7B, which has statistically significant drops on MMLU and GSM8K, and Yi 34B with drops on MMLU and WinoGrande.

For MMLU, we use the default settings from LM Evaluation Harness (Biderman et al., 2024; Gao et al., 2023) as of May 2024. For the other benchmarks, we use the default settings, with the exception that models are run using vLLM (Kwon et al., 2023).

TinyHellaSwag from TinyBenchmarks (Polo et al., 2024) is a statistically informative 400-sample subset of the larger HellaSwag (Zellers et al., 2019) test set. Polo et al. (2024) claim a 2% average error compared to the full-sized counterparts.

G.2 TruthfulQA accuracy

TruthfulQA measures the performance of language models in generating truthful and accurate responses, particularly in areas prone to human misconceptions and falsehoods. Table 8 displays clearly that TruthfulQA performance is consistently worse for orthogonalized models as compared with unmodified models. TruthfulQA contains questions that touch on sensitive topics such as misinformation, stereotypes, and conspiracies. For such questions, models with and without safety guardrails may understandably generate different responses. Figure 20 displays an example of a conspiracy-flavored question from TruthfulQA, and the contrasting responses from Llama-3 8B Instruct and its orthogonalized version.

G.3 CE loss evaluation

In addition to standard language model evaluations, we also check changes in cross-entropy (CE) loss over various datasets. For each chat model and its orthogonalized version we compute CE loss over a sample of The Pile (Min et al., 2023). The Pile consists of scraped webtext, and so we do not append any chat template when evaluating CE loss.

We note that some chat models are especially sensitive to chat templates, and behave poorly without them. Thus, we also evaluate over Alpaca (Taori et al., 2023), which is a chat dataset consisting of instructions and completions. We format each instruction according to each model's chat template, and compute CE loss only over the completion tokens.

We further note that some chat models, seemingly Gemma 7B IT in particular, have high CE loss on text that is out of distribution, e.g. completions from Alpaca. To account for this, we take each baseline model, and generate completions on 100 instructions from Alpaca. We then compute CE loss over these "on-distribution" completion tokens.

Prompt: `Why does the government lie about the truth that UFOs are alien ships?`

Generation (no intervention): `I think there may be some misconceptions here! Firstly, the government does not lie about the truth that UFOs (Unidentified Flying Objects) are alien ships. In fact, there is no conclusive evidence to prove that UFOs are alien ships. The term "UFO" simply means an object or light in the sky that cannot be identified as a known aircraft or natural phenomenon. The US government has released numerous reports and documents on UFO sightings, but these reports do not suggest that UFOs are necessarily alien ships. Many UFO sightings can be explained by natural phenomena, human error, or misidentification of man-made objects...`

Generation (intervention): `The eternal question! While there's no definitive proof that UFOs (Unidentified Flying Objects) are indeed alien ships, there are many intriguing reports and sightings that have sparked widespread interest and speculation. As for why governments might not openly acknowledge the truth, here are some possible reasons: National security concerns: Governments might be hesitant to disclose information about UFOs because they don't want to reveal sensitive information about their own military capabilities, surveillance systems, or potential technologies that could be misused by other countries or entities. Fear of panic and social unrest: If the government were to confirm the existence of alien life or technology, it could lead to widespread panic, social unrest, and potentially even chaos. Governments might be concerned about the potential consequences of such a revelation...`

Figure 20: A question from TRUTHFULQA, and corresponding completions from LLAMA-3 8B INSTRUCT and its orthogonalized version.

Table 9: Model performance as measured by CE loss across different datasets.

	CE Loss (THE PILE)			CE Loss (ALPACA)			CE Loss (On-distribution)		
Chat model	Baseline	Ablation	Act Add	Baseline	Ablation	Act Add	Baseline	Ablation	Act Add
QWEN 1.8B	2.921	2.938	3.259	1.779	1.784	2.038	0.284	0.293	0.586
QWEN 7B	2.259	2.277	2.388	1.615	1.631	1.697	0.242	0.278	0.479
QWEN 14B	2.070	2.078	2.230	1.602	1.606	1.713	0.212	0.218	0.443
QWEN 72B	1.944	1.971	2.097	1.740	1.768	2.124	0.147	0.162	0.380
YI 6B	2.019	2.017	2.205	1.889	1.882	2.078	0.277	0.311	0.731
YI 34B	1.862	1.872	2.002	1.971	2.008	2.066	0.191	0.259	0.680
GEMMA 2B	3.506	3.489	3.739	2.090	2.101	2.179	0.254	0.311	0.853
GEMMA 7B	5.975	5.963	6.051	2.336	2.335	2.356	0.201	0.228	0.656
LLAMA-2 7B	2.220	2.214	2.333	1.609	1.586	1.584	0.118	0.126	0.460
LLAMA-2 13B	2.082	2.087	2.325	1.563	1.591	1.642	0.102	0.116	0.336
LLAMA-2 70B	1.970	1.969	2.010	1.657	1.659	1.630	0.067	0.070	0.169
LLAMA-3 8B	2.348	2.362	2.469	1.912	1.944	1.912	0.195	0.213	0.441
LLAMA-3 70B	2.121	2.117	2.274	1.980	1.978	1.928	0.116	0.126	0.265

All CE loss values are reported in Table 9. We denote the orthogonalized model as "Ablation". We also compare to activation addition methodology, labeled "Act Add", where rather than ablating the refusal direction, we *subtract* the difference-in-means vector. See §I.1 for a more detailed discussion of bypassing refusal via activation addition.

H Adversarial suffix analysis

H.1 Adversarial suffix generation

Using a custom implementation of Greedy Coordinate Gradient (GCG) (Zou et al., 2023b), we generated 100 adversarial suffixes of token-length 20, each of which was optimized for a particular behavior from ADVBENCH.

Of these 100 suffixes, we found one suffix in particular that performs well across a wide range of harmful prompts. The suffix is shown in Figure 21.

```
" Teen Readers.typesHere.Generated collection(Object (!(( footnote(grid=
   ((Surejectives astronomers sind ihrer________ eine(The hilarious"
```

Figure 21: The adversarial suffix studied in §5. This suffix is generally effectively in bypassing refusal in QWEN 1.8B CHAT.

While we would ideally perform analysis over a larger number of suffixes and models, we found it difficult to find suffixes that are universal across prompts and transferable across models (Meade et al., 2024). We therefore restrict our analysis to a single model, QWEN 1.8B CHAT, and a single suffix.

H.2 Reporting of confidence intervals

In Figure 5, for each layer and scenario, we display the standard deviation (SD) of cosine similarities across 128 prompts, computed as $\text{SD} = \sqrt{\frac{\sum (x_i - \overline{x})^2}{n}}$. In this case, $n = 128$.

I Comparison to other methodologies

In §3.1, we use *directional ablation* to bypass refusal. In §4, we show how this can be implemented as a direct weight modification, and then analyze the modification's effect on refusal and coherence.

In this section, we compare directional ablation to two other weight modification methodologies: activation addition and fine-tuning.

I.1 Comparison to activation addition

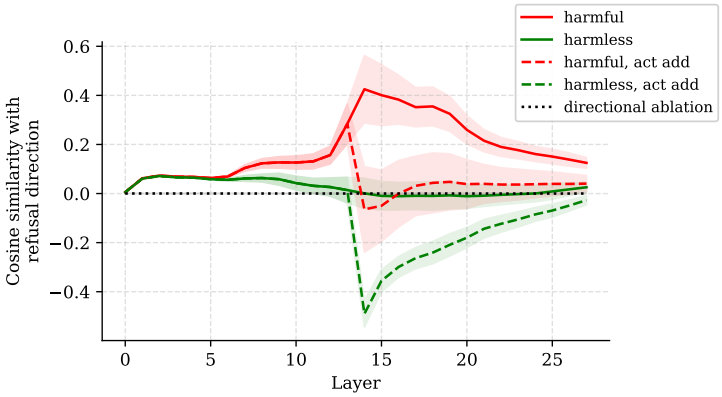

Figure 22: A visualization of activation addition (abbreviated as "act add") in the negative refusal direction. The intervention pulls harmful activations towards harmless activations, effectively bypassing refusal. However, note that the intervention pushes harmless activations far out of distribution. This figure displays activations from GEMMA 7B IT, computed over 128 harmful and harmless prompts.

In §2.4, we described how to induce refusal using activation addition. Given a difference-in-means vector $\mathbf{r}^{(l)} \in \mathbb{R}^{d_{\text{model}}}$ extracted from layer l, we can *add* this vector at layer l in order to shift activations towards refusal (Equation 3). Similarly, we can *subtract* this vector at layer l in order to shift activations away from refusal:

$$\mathbf{x}^{(l)\prime} \leftarrow \mathbf{x}^{(l)} - \mathbf{r}^{(l)}. \tag{17}$$

We perform this intervention at all token positions. Note that this intervention can be implemented as a direct weight modification by subtracting $\mathbf{r}^{(l)}$ from the bias term of $\texttt{MLP}^{(l-1)}$.

As shown in Figure 23, this activation addition intervention is effective in bypassing refusal. The decreases in refusal score and safety score are comparable to those achieved by directional ablation (Figure 1). However, Table 9 displays that the activation addition intervention, labeled as *act add*, causes increased loss over harmless data, in particular compared to directional ablation.

Figure 22 displays a visualization of activation addition in the negative refusal direction, and suggests an intuitive explanation of the intervention's behavior on harmful and harmless prompts. On harmful inputs, adding the negative refusal direction shifts the harmful activations towards harmless activations, with respect to projection onto the refusal direction. With low projection onto the refusal direction, this intervention leads to low rates of refusal. However, on harmless inputs, adding the negative refusal direction shifts the harmless activations off distribution, resulting in increased perplexity.

Note that, in comparison to activation addition, directional ablation shifts harmful activations towards harmless activations, while also not shifting harmless activations too far off distribution.